

RES-VER — Verification, Testing and Uncertainty Methodology

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1 Domain Purpose and Scope

2 Domain Deliverables

3 RES-VER-FRM — Simulation Credibility and Test Framework

3.1 Purpose and Scope

This Deliverable defines the validation hierarchy for the local three-dimensional URANS–VOF tsunami–barrier model. It separates solver capability, event-conditioned hydrodynamics and local impact/loading credibility.

3.2 Research Questions

3.3 Terminology and Definitions

3.4 Literature Evidence

Qin et al. [1] is retained as evidence that local model validation must include velocity and force, not only free-surface height. Watanabe et al. [2] is retained as evidence that blind tsunami-impact simulations can vary materially in force and velocity predictions even when the general hydrodynamic setup is shared. Source role: ‘3D-VALIDATION’. Project conclusion: local validation shall be staged and multi-output. Limitation: passing one stage does not automatically validate the next stage.

3.5 Governing Relations and Quantitative Definitions

3.6 Test Objective

The local validation objective is:

V1 solver capability → V2 event-conditioned hydrodynamic realism → V3 local impact and loading credibility

V1 checks that the numerical method can solve canonical free-surface and impact problems. V2 checks that the selected event and local forcing reproduce relevant hydrodynamic observations. V3 checks that pressure, force, impulse, moment, overtopping and transmitted/reflected wave outputs are credible for barrier-performance assessment.

3.7 Reference or Benchmark Solution

3.8 Test Configuration

3.9 Measured Quantities

The framework shall measure free-surface elevation, arrival time, velocity, run-up, overtopping, phase volume, pressure, shear, resultant force, force impulse, moment, transmission/reflection measures and computational diagnostics.

3.10 Error Metrics

3.11 Tolerance and Acceptance Criteria

Acceptance shall be stage-specific. A model that passes V1 remains unvalidated for the selected event until V2 passes. A model that passes V2 remains unvalidated for barrier loading until V3 passes. URANS $k-\omega$ SST shall be accepted for screening only where it preserves the safety-critical quantities and candidate ranking within the tolerances defined in ‘RES-VER-MET’.

3.12 Execution Handoff

Research deliverables specify the credibility hierarchy and required metrics only. Software validation reports shall state whether a result is V1, V2 or V3 evidence, which dataset and geometry identifiers were used, and which acceptance criteria passed. No validation shall be claimed from the existence of this specification alone.

3.13 Candidate Approaches

3.14 Critical Comparison

3.15 Assumptions and Applicability

3.16 Limitations and Failure Conditions

3.17 Project-Specific Conclusions

3.18 Selected Approach and Rationale

The selected credibility framework is the three-stage V1–V2–V3 hierarchy. It prevents water-level agreement, solver stability or successful regional propagation from being treated as sufficient validation of local pressure and loading.

3.19 Unresolved Decisions

3.20 Requirements and Handoffs

‘RES-VER-BMK’ shall define benchmark classes for V1. ‘RES-VER-MET’ shall define output-specific metrics and tolerances for V2 and V3. Software validation reports shall identify which stage each result supports.

3.21 Deliverable Completion Record

4 RES-VER-BMK — Analytical and Canonical Benchmark Suite

4.1 Purpose and Scope

4.2 Research Questions

4.3 Terminology and Definitions

4.4 Literature Evidence

4.4.1 Grilli et al. (2013)

Grilli et al. [3] constructed a complete Tōhoku hindcast linking a transient earthquake source, non-hydrostatic tsunami generation, dispersive regional propagation, and nested coastal inundation. The model was compared with near- and far-field deep-ocean records, offshore GPS-wave gauges, run-up measurements, and inundation observations.

The paper is retained as evidence for a stage-specific benchmark in which source generation, offshore propagation, coastal transformation, and inundation are assessed separately.

4.4.2 Wei et al. (2013)

Wei et al. [4] used a source constrained by two deep-ocean tsunami records to drive eleven nested inundation models along the Japanese coastline. The resulting simulations were compared with independent offshore GPS records, nearshore gauges, surveyed heights, and mapped inundation areas.

The paper is retained as evidence that one regional source and propagation solution can supply several local models, and that observations used to infer the source must be distinguished from those used to assess its predictive performance.

4.5 Governing Relations and Quantitative Definitions

4.6 Test Objective

The benchmark suite shall verify solver capability before event validation and loading validation are attempted. Local 3D benchmarks shall isolate free-surface transport, pressure correction, wave generation, wall loading and overtopping before the full event-conditioned barrier model is used.

4.7 Reference or Benchmark Solution

The Tōhoku benchmark shall be divided into the following validation stages:

1. earthquake-source and seabed-deformation assessment;
2. initial tsunami-generation assessment;
3. near-source offshore propagation;
4. regional propagation towards the selected coastline;
5. nearshore transformation;
6. run-up and inundation;
7. local barrier interaction.

Failure at one stage shall not be concealed by acceptable agreement at a later stage. For example, an incorrect source shall not be accepted solely on the basis of matching one inundation contour after roughness or bathymetry has been adjusted.

The local V1 solver-capability benchmark classes are:

1. still water with gravity and a flat free surface;
2. VOF advection of a prescribed interface;
3. dam-break or bore propagation;
4. regular or solitary wave generation and absorption in a numerical wave tank;
5. solitary-wave run-up on a slope;
6. wave impact on a vertical wall or simple fixed obstacle;
7. overtopping of a simple impermeable barrier;
8. flow around a fixed object with pressure and force extraction;
9. mesh and timestep refinement for the local quantities of interest.

Watanabe et al. [2] and Qin et al. [1] support the separation between hydrodynamic validation and impact/load validation. Source role: ‘3D-VALIDATION’. Project conclusion: pressure, velocity and force benchmarks are required before local barrier-performance claims are accepted.

4.8 Test Configuration

4.9 Simulation Duration

The benchmark interval shall include all waves that materially affect the selected coastal and structural quantities.

The local forcing history shall not be reduced automatically to:

- the first arriving crest;
- the maximum isolated amplitude;
- one idealised solitary wave;
- a short interval surrounding initial impact.

Later waves may propagate over residual inundation water, interact with coastal reflections, and produce the maximum inundation, overtopping, or cumulative load after the first arrival.

4.10 Measured Quantities

Local benchmark outputs shall include phase-volume conservation, free-surface elevation, wave-front arrival, depth-dependent velocity, run-up, overtopping volume, pressure histories, pressure distribution, resultant force, impulse, moment and reflected/transmitted wave measures.

4.11 Error Metrics

4.12 Tolerance and Acceptance Criteria

4.13 Execution Handoff

4.14 Candidate Approaches

4.15 Critical Comparison

4.16 Assumptions and Applicability

4.17 Limitations and Failure Conditions

4.18 Project-Specific Conclusions

4.19 Selected Approach and Rationale

The selected benchmark approach is progressive. A full event-conditioned local barrier simulation shall not be the first test of the 3D solver; simpler canonical cases shall verify each numerical and physical mechanism before combined validation.

4.20 Unresolved Decisions

4.21 Requirements and Handoffs

Software benchmark reports shall identify the benchmark class, geometry, mesh, timestep limits, boundary conditions, turbulence mode, measured quantities and pass/fail criteria. ‘RES-VER-MET’ shall define the numerical values of the acceptance metrics.

4.22 Deliverable Completion Record

5 RES-VER-TST — Component, Conservation and Interface Test Design

5.1 Purpose and Scope

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.4.1 Qin et al. (2018)

Qin et al. [1] demonstrated that a local three-dimensional model should be assessed against depth-dependent velocity and direct structural-force measurements rather than only against free-surface elevation.

5.4.2 Watanabe et al. (2022)

Watanabe et al. [2] compared multiple numerical predictions against withheld laboratory measurements. The blind comparison showed that three-dimensional models could reproduce non-hydrostatic and short-wave behaviour but also produced substantial variability in velocity and force. Smaller cells did not systematically reduce this variability.

5.5 Local Tsunami-Impact Test Sequence

The selected parent equations shall be validated as part of a complete modelling workflow. Agreement in water level shall not be treated as evidence of accurate pressure, force, or structural loading.

The local model shall be assessed progressively using:

1. incident-wave generation without a structure;

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2. free propagation and transformation within the local domain;
 3. impact on an isolated impermeable structure;
 4. pressure and force on a simple canonical geometry;
 5. overtopping and run-up on a barrier;
 6. separation, wake, and downstream-flow validation;
 7. interaction with multiple or finite-width structures;
 8. one-way structural load transfer;
 9. two-way FSI where activated.

Each stage shall isolate a limited set of physical and numerical mechanisms before additional complexity is introduced.

5.6 One-Way and Simultaneously Coupled Model Comparison

5.6.1 Test Objective

The test shall determine whether neglecting feedback from the local three-dimensional barrier domain produces a material change in the quantities used to assess candidate barrier performance. The comparison shall distinguish:

1. the one-way replay model, in which a saved two-dimensional state drives the local three-dimensional model without reverse feedback;
2. the simultaneously coupled model, in which the retained two-dimensional buffer and local three-dimensional model exchange information during the impact calculation.

5.6.2 Test Configuration

Both branches shall begin from the same immutable pre-impact checkpoint and shall use identical:

- environmental scenario;
- barrier geometry;
- material and boundary assumptions;
- interface location;
- two-dimensional and three-dimensional meshes;
- physical simulation interval;
- output locations;

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- numerical tolerances;
 - quantity definitions.

The comparison shall be repeated for at least:

- the reference barrier;
- a weakly reflecting candidate;
- a strongly reflecting or obstructive candidate;
- more than one interface location.

5.6.3 Concurrent Execution

The two branches may be launched concurrently to reduce elapsed comparison time. Each branch shall receive a separate copy of the saved checkpoint and separate mutable simulation state. A synchronisation barrier shall prevent the comparison and acceptance logic from executing until both simulations have completed successfully. A failed branch shall invalidate the comparison rather than permitting the surviving result to be accepted in isolation. The assigned cores, threads, accelerators, and memory shall be recorded. Concurrent execution shall not oversubscribe the available hardware or change the numerical configuration relative to separately executed reference runs.

5.6.4 Measured Quantities

The comparison shall include:

- free-surface elevation upstream and downstream of the barrier;
- reflected-wave amplitude and energy;
- transmitted wave energy;
- normal mass flux across the interface;
- momentum flux across the interface;
- peak barrier force;
- barrier-force impulse;
- peak pressure;
- overtopping volume;
- downstream maximum depth and velocity;
- regional run-up or inundation where affected;
- wall-clock time, processor time, and memory consumption.

5.6.5 Error Metrics

For a scalar quantity of interest q , the relative difference shall be evaluated using:

$$\delta_q = \frac{|q_{\text{coupled}} - q_{\text{one-way}}|}{\max(|q_{\text{coupled}}|, q_{\text{scale}})} \quad (5.1)$$

where q_{scale} is a non-zero reference scale used to avoid unstable normalisation for quantities close to zero. For a time-dependent quantity $q(t)$, the normalised difference may be evaluated using:

$$\delta_{q,2} = \frac{\|q_{\text{coupled}} - q_{\text{one-way}}\|_{L^2}}{\max(\|q_{\text{coupled}}\|_{L^2}, q_{\text{scale}})} \quad (5.2)$$

Peak-value, phase, impulse, and spatial-location errors shall be reported separately rather than being concealed within one aggregate norm.

5.6.6 Tolerance and Acceptance Criteria

The one-way replay model may be accepted for broad candidate screening when:

- all safety-critical and optimisation quantities remain within their prescribed fidelity tolerances;
- interface mass and momentum imbalances remain acceptable;
- reflected-wave differences do not materially alter the upstream or downstream response;
- the conclusion remains stable under movement of the interface;
- the ranking of candidate geometries is unchanged.

The simultaneously coupled model shall be required when any safety-critical quantity exceeds its tolerance, when candidate rankings change, or when barrier-generated feedback materially modifies the regional response. The acceptance tolerances shall be defined after the characteristic magnitudes, numerical errors, and decision sensitivity of each quantity have been established. They shall not be selected solely to favour the cheaper model.

5.6.7 Execution Handoff

The Software workstream shall automate:

1. restoration of the common checkpoint;
2. creation of isolated branch states;

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3. concurrent or sequential execution;
 4. branch-completion synchronisation;
 5. failure detection;
 6. metric extraction;
 7. comparison against the configured tolerances;
 8. generation of a reproducible test report.

The Optimisation workstream shall use the approved fidelity policy to select the solver mode for candidate screening and final confirmation.

5.7 Interface-Location Sensitivity Test

5.7.1 Test Objective

The test shall determine whether the selected two-dimensional–three-dimensional interfaces lie within regions where the assumptions of their receiving models are valid.

Separate studies shall be performed for:

1. the upstream two-dimensional-to-three-dimensional lifting interface;
2. the downstream three-dimensional-to-two-dimensional restriction interface;
3. the bidirectional interface used by the simultaneously coupled configuration.

5.7.2 Test Configuration

At least three candidate interface positions shall be examined:

1. a position close to the local barrier-interaction region;
2. a position near the expected transition in model validity;
3. a position farther into the region where the receiving model is expected to be valid.

The candidate locations shall be selected with reference to:

- bathymetric and topographic gradients;
- geometric contractions and expansions;
- predicted breaking and overtopping regions;
- vertical velocity and shear;

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- free-surface fragmentation;
 - recirculation and wake length;
 - reflected-wave propagation;
 - wetting–drying behaviour;
 - proximity to artificial boundaries.

All other physical and numerical inputs shall remain unchanged between the interface-location cases.

5.7.3 Measured Quantities

The test shall compare:

- free-surface time histories;
- depth-integrated discharge;
- incident, reflected, and transmitted wave amplitudes;
- incident, reflected, and transmitted energy;
- peak barrier force;
- barrier-force impulse;
- peak pressure and pressure impulse;
- overtopping volume;
- downstream depth and velocity;
- run-up and inundation extent;
- interface mass imbalance;
- interface momentum imbalance;
- amplitude and phase spectra.

5.7.4 Error Metrics

For a quantity of interest q evaluated using neighbouring interface positions Γ_i and Γ_{i+1} , the relative location sensitivity shall be defined as:

$$\delta_{\Gamma,q}^{(i)} = \frac{|q_{\Gamma_{i+1}} - q_{\Gamma_i}|}{\max(|q_{\Gamma_{i+1}}|, q_{\text{scale}})} \quad (5.3)$$

where q_{scale} is a non-zero characteristic scale.

For a transferred waveform $\eta(t)$, the amplitude spectrum shall be evaluated using:

$$\hat{\eta}(f) = \mathcal{F} \{ \eta(t) \} \quad (5.4)$$

Spectral amplitude and phase differences shall be reported alongside time-domain norms.

5.7.5 Tolerance and Acceptance Criteria

An interface position may be accepted when:

- moving the interface farther into the receiving model's valid region produces no material change in the engineering quantities of interest;
- mass and momentum transfer errors remain within their prescribed tolerances;
- the transferred waveform retains the required amplitude, phase, and frequency content;
- candidate-design rankings remain unchanged;
- no material artificial-boundary contamination is detected;
- the receiving model reproduces the donor behaviour over an overlapping comparison region.

The cheapest acceptable interface position shall be selected. A location shall not be chosen solely to minimise the size of the three-dimensional domain.

5.7.6 Execution Handoff

The Software workstream shall permit interface locations to be changed without modifying either governing solver. Mesh generation, mapping, checkpoint extraction, and test execution shall be parameterised by the interface definition.

The verification software shall execute all candidate interface locations, wait until every required branch has completed, and generate a comparative report containing field, flux, spectral, and engineering metrics.

5.8 Governing Relations and Quantitative Definitions

5.9 Test Objective

5.10 Reference or Benchmark Solution

5.11 Test Configuration

5.12 Measured Quantities

5.13 Error Metrics

5.14 Tolerance and Acceptance Criteria

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5.20 Project-Specific Conclusions

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5.22 Unresolved Decisions

5.23 Requirements and Handoffs

5.24 Deliverable Completion Record

6 RES-VER-CNV — Convergence and Numerical-Error Test Methodology

6.1 Purpose and Scope

6.2 Research Questions

6.3 Terminology and Definitions

6.4 Literature Evidence

6.5 Governing Relations and Quantitative Definitions

-
1. spatial and temporal discretisation sensitivity;
 2. incident-wave and boundary-condition sensitivity;
 3. turbulence-closure sensitivity;
 4. wall-treatment sensitivity;
 5. free-surface-capturing sensitivity;
 6. pressure-filtering and load-integration sensitivity;
 7. fluid-to-structure mapping sensitivity;
 8. structural time-integration and damping sensitivity.

Grid refinement shall not be treated as sufficient evidence of model validity. A refined solution may converge towards an incorrect result where the boundary conditions, closure, constitutive model, or incident state are inappropriate.

6.7 Reference or Benchmark Solution

6.8 Test Configuration

6.8.1 Spatial Resolution Requirements

Grid adequacy shall be determined for each output separately.

A grid that reproduces a deep-ocean leading-wave amplitude shall not be assumed adequate for:

- narrow bays;
- steep ria coastlines;
- local run-up;
- river propagation;
- coastal structures;
- barrier-scale pressure and force.

Computational-grid spacing and source-data resolution shall be reported independently. Refinement of the computational mesh below the resolution of the bathymetric or topographic input shall not be treated as equivalent physical refinement.

6.9 Measured Quantities

6.10 Error Metrics

6.11 Tolerance and Acceptance Criteria

6.12 Execution Handoff

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6.21 Deliverable Completion Record

7 RES-VER-MET — Test Metrics, Tolerances and Acceptance Criteria

7.1 Purpose and Scope

7.2 Research Questions

7.3 Terminology and Definitions

7.4 Literature Evidence

Qin et al. [1] supports validation against depth-dependent velocity and direct force extraction. Watanabe et al. [2] supports multi-output acceptance because water level, velocity and force can exhibit different error behaviour. Source role: ‘3D-VALIDATION’. Project conclusion: no single scalar error metric is sufficient for accepting the local URANS–VOF model.

7.5 Governing Relations and Quantitative Definitions

7.6 Test Objective

7.7 Reference or Benchmark Solution

7.8 Test Configuration

7.9 Measured Quantities

7.9.1 Local-Impact Validation Observables

The minimum validation set shall include:

- phase-volume and mass-conservation diagnostics;
- α boundedness and interface-compression diagnostics;
- maximum local CFL and rejected-timestep count;
- pressure-correction residual and pressure-corrector count;
- free-surface elevation at multiple upstream and downstream locations;
- wave-front arrival time;
- depth-dependent velocity;
- inundation depth;
- overtopping volume or discharge;
- local pressure histories at several elevations;
- spatial pressure distribution;
- resultant force;
- resultant moment and force/moment history;
- force impulse;
- overturning moment;
- energy, reflection and transmission metrics;
- transmitted and reflected wave measures;
- wake and recirculation characteristics;
- structural displacement and restoring force where FSI is active.

The maximum value and complete time history shall both be assessed. A model that reproduces one peak while misrepresenting load duration, phase, or impulse shall not be accepted without further diagnosis.

7.9.2 Benchmark Validation Quantities

The benchmark shall report separately:

- first-arrival time;
- leading-crest and leading-trough amplitude;
- waveform phase;
- dominant period and spectral content;
- maximum surface elevation;
- depth-integrated discharge or velocity;
- inundation depth;
- inundation extent;
- maximum run-up;
- reflected-wave behaviour;
- complete-wave-train duration.

A time-shifted waveform comparison may be used to assess waveform shape, but the unshifted arrival-time error shall be reported first. An imposed time shift shall not be treated as a correction to the physical simulation.

7.10 Error Metrics

7.10.1 Free-Surface and Conservation Measures

For free-surface elevation η , the normalised free-surface error is:

$$E_\eta = \frac{\|\eta_{\text{sim}} - \eta_{\text{obs}}\|_2}{\|\eta_{\text{obs}}\|_2 + \epsilon_\eta} \quad (7.1)$$

The local mass-conservation acceptance metric is:

$$E_{\text{mass}} = \max \left(\max_i \hat{R}_{m,i}, |E_m| \right) \quad (7.2)$$

where $\hat{R}_{m,i}$ and E_m are defined by the normalised local mass residual and global mass-imbalance diagnostics in ‘RES-NUM-STB’ and ‘RES-NUM-ERR’.

VOF boundedness shall be reported as:

$$E_{\alpha, \text{bound}} = \max_i [\max(0, -\alpha_i, \alpha_i - 1)] \quad (7.3)$$

The maximum accepted-timestep Courant diagnostic shall be:

$$\text{CFL}_{\max} = \max_f \left(\frac{|\phi_f| \Delta t}{V_{c(f)} + \epsilon_V} \right) \quad (7.4)$$

The pressure-correction residual shall be normalised as:

$$E_p = \frac{\|\mathbf{A}_p p' - \mathbf{b}_p\|_2}{\|\mathbf{b}_p\|_2 + \epsilon_p} \quad (7.5)$$

Decision role: these diagnostics are software-emitted acceptance metrics for the RES-NUM-SPEC local solving procedure. Supporting evidence: Watanabe et al. [2] and Qin et al. [1] support multi-output local validation; the numerical residual and CFL definitions are project acceptance diagnostics tied to the selected pressure-correction/VOF procedure. Limitation: final tolerance values remain open.

7.10.2 Local-Impact Error Measures

For a measured quantity $q_{\text{obs}}(t)$ and simulated quantity $q_{\text{sim}}(t)$, the time-series root-mean-square error is:

$$\text{RMSE}_q = \sqrt{\frac{1}{N} \sum_{n=1}^N [q_{\text{sim}}(t_n) - q_{\text{obs}}(t_n)]^2} \quad (7.6)$$

The peak-value ratio is:

$$R_{q, \max} = \frac{\max_t |q_{\text{sim}}(t)|}{\max_t |q_{\text{obs}}(t)|} \quad (7.7)$$

The normalised impulse error is:

$$E_J = \frac{\|\mathbf{J}_{\text{sim}} - \mathbf{J}_{\text{obs}}\|}{\|\mathbf{J}_{\text{obs}}\| + \epsilon_J} \quad (7.8)$$

The normalised moment error is:

$$E_M = \frac{\|\mathbf{M}_{\text{sim}} - \mathbf{M}_{\text{obs}}\|}{\|\mathbf{M}_{\text{obs}}\| + \epsilon_M} \quad (7.9)$$

The overtopping-volume error is:

$$E_{\text{ot}} = \frac{|V_{\text{ot,sim}} - V_{\text{ot,obs}}|}{|V_{\text{ot,obs}}| + \epsilon_{\text{ot}}} \quad (7.10)$$

The run-up error is:

$$E_{\text{ru}} = \frac{|R_{\text{u,sim}} - R_{\text{u,obs}}|}{|R_{\text{u,obs}}| + \epsilon_{\text{ru}}} \quad (7.11)$$

The energy-diagnostic error for any selected energy metric $E_\star$ is:

$$E_{\text{energy},\star} = \frac{|E_{\star,\text{sim}} - E_{\star,\text{ref}}|}{|E_{\star,\text{ref}}| + \epsilon_\star} \quad (7.12)$$

Source role: this metric provides a common acceptance form for the incident, reflected, transmitted and apparent-dissipation measures in RES-PHY-BAR. Supporting evidence: Watanabe et al. [2] supports separate metric reporting rather than one aggregate score. Limitation: the reference energy measure may be experimental, high-fidelity numerical or comparative depending on the V1–V2–V3 stage.

Force and moment extraction shall also pass surface-integration consistency checks by comparing integrated patch traction with component pressure and shear contributions:

$$E_{\text{F,int}} = \frac{\|\mathbf{F} - \mathbf{F}_p - \mathbf{F}_\tau\|}{\|\mathbf{F}\| + \epsilon_F} \quad (7.13)$$

7.11 Tolerance and Acceptance Criteria

7.11.1 Local-Model Acceptance Principle

The operational URANS model shall be accepted only where it satisfies both:

1. experimental validation requirements for the quantities relevant to barrier performance and structural loading;
2. controlled agreement with the selected high-fidelity model for representative candidate geometries.

Neither URANS nor LES shall be accepted solely from its theoretical fidelity. The selected formulation must demonstrate that it preserves:

- the relevant physical mechanisms;
- the required engineering outputs;
- the relative performance ranking of candidate barriers;
- conservative and reproducible load transfer to the structural model.

The final acceptance tolerances shall account for experimental uncertainty, measurement filtering, repeatability, and the intended use of each model-fidelity level.

Acceptance for the baseline local 3D model shall include:

- free-surface elevation and arrival-time error below tolerance;
- VOF boundedness, phase volume and mass-conservation diagnostics below tolerance;
- CFL, pressure residual and rejected-timestep counts below tolerance;
- pressure, force-history, impulse and moment-history errors below tolerance;
- run-up and overtopping errors below tolerance;
- incident, reflected, transmitted and dissipated-energy metrics reported and below the stage-appropriate tolerance where reference data exist;
- no unresolved boundary-reflection contamination;
- stable candidate ranking under mesh, timestep and URANS–LES comparison studies.

7.12 Execution Handoff

7.13 Candidate Approaches

7.14 Critical Comparison

7.15 Assumptions and Applicability

7.16 Limitations and Failure Conditions

7.17 Project-Specific Conclusions

The local 3D model is valid as a baseline high-Reynolds-number hydrodynamic tsunami-barrier interaction model for run-up, overtopping, free-surface deformation, pressure, force, moment, and comparative barrier-performance analysis. It is not valid yet for structural failure, scour, deformable barriers, detailed aeration, sediment transport, or final design certification.

7.18 Selected Approach and Rationale

The selected acceptance approach is output-specific. Free-surface, mass-conservation, pressure, force, impulse, moment, run-up and overtopping metrics shall be reported separately rather than collapsed into one aggregate validation score.

7.19 Unresolved Decisions

Open items are final numerical tolerance values, observation datasets, experimental uncertainty models, acceptable URANS–LES discrepancy, ranking-stability threshold and certification-level design criteria.

7.20 Requirements and Handoffs

Software validation reports shall provide the metrics in this Deliverable for each accepted local run, including mass conservation, α boundedness, CFL, pressure residual, force/moment histories, run-up, overtopping and energy metrics. ‘RES-ML-DAT’ shall use only validated outputs for any future training data, and ‘RES-OPT-OBJ’ shall not treat unvalidated hydrodynamic metrics as final design objectives.

7.21 Deliverable Completion Record

8 RES-VER-UQ — Sensitivity and Uncertainty Assessment Methodology

8.1 Purpose and Scope

8.2 Research Questions

8.3 Terminology and Definitions

8.4 Literature Evidence

8.5 Governing Relations and Quantitative Definitions

8.6 Test Objective

8.7 Reference or Benchmark Solution

8.8 Test Configuration

8.9 Measured Quantities

8.10 Error Metrics

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face. It defines required tests and emitted metrics only; it does not claim that validation has been executed.

9.2 Research Questions

9.3 Terminology and Definitions

9.4 Literature Evidence

Grilli et al. [3] and Wei et al. [4] support a Tohoku hindcast stage in which source, propagation and coastal hydrodynamic observations are compared before local impact claims are made. Qin et al. [1] and Watanabe et al. [2] support local validation using velocity, pressure and force outputs rather than free-surface elevation alone. Project conclusion: the test specification shall be staged as V1 solver capability, V2 Tohoku hydrodynamics and V3 local impact/loading.

9.5 Governing Relations and Quantitative Definitions

9.6 Test Objective

The integrated test objective is to verify that:

1. V1 solver-capability tests pass for regional propagation, local free-surface flow, wall/obstacle impact and conservation;
2. V2 Tohoku hydrodynamic tests compare source-to-coast propagation, arrival time, waveform, run-up and inundation against accepted observations for the Kamaishi and Sendai corridors;
3. V3 impact/loading tests compare local pressure, force, moment, impulse, overtopping and energy outputs for fixed rigid barrier classes.

The locked 2D NLSWE finite-volume specification must preserve conservation, positivity, well-balanced still water, moving-bed source behaviour, wet/dry shoreline motion, boundary non-reflection and Tohoku-scale propagation before it can drive accepted replay histories.

9.7 Reference or Benchmark Solution

9.8 Test Configuration

9.9 Measured Quantities

Regional measured quantities shall include:

- water depth h and free-surface elevation η ;

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- depth-integrated discharges q_x and q_y ;
 - arrival time, peak elevation, phase and dominant period;
 - run-up elevation, inundation distance and wet/dry extent;
 - mass balance and momentum-balance diagnostics;
 - boundary-reflection indicators and sponge-layer attenuation;
 - mesh-convergence and CFL-sensitivity measures.

Local and interface measured quantities shall include:

- α boundedness, phase volume and mass conservation;
- local CFL, pressure-correction residual and rejected timestep count;
- replay-boundary water level, velocity, discharge and pressure state;
- local pressure and shear histories on barrier patches;
- resultant force and moment histories;
- force impulse, pressure impulse and centre of pressure;
- run-up, overtopping and transmitted/reflected energy metrics.

9.10 Error Metrics

9.11 Tolerance and Acceptance Criteria

9.12 Execution Handoff

The following V1–V2–V3 verification and validation sequence is required before validation is declared complete:

1. V1 regional lake at rest over non-flat bathymetry;
2. V1 one- and two-dimensional dam-break or bore propagation;
3. V1 wet/dry run-up with positivity and mass-balance audit;
4. V1 local VOF boundedness, pressure-correction residual and wall/obstacle impact benchmarks;
5. V1 moving-bottom tsunami-generation test;
6. V1 open/radiation boundary and sponge-reflection test;
7. V1 mesh refinement and CFL sensitivity on regional and local domains;

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8. V2 Tohoku waveform/current/run-up or inundation comparison using data not already consumed by source calibration.
 9. V2 Kamaishi and Sendai replay-boundary consistency checks.
 10. V3 local fixed-barrier run-up, overtopping, pressure, force, moment, impulse and energy-metric tests.

Execution evidence belongs to the verification and Software workstreams and shall not be inferred merely from completion of the research specification.

9.13 Candidate Approaches

9.14 Critical Comparison

9.15 Assumptions and Applicability

9.16 Limitations and Failure Conditions

9.17 Project-Specific Conclusions

The regional mathematical and numerical model is ready for verification planning, and the local fixed-rigid barrier model now has a staged acceptance path. Neither is yet a validated solver or validated Tohoku case-study simulation.

9.18 Selected Approach and Rationale

9.19 Unresolved Decisions

Open verification decisions include final benchmark tolerances, final case-study observation sets, calibration/validation split, final mesh refinement levels, and acceptance thresholds for boundary reflection and wet/dry mass-balance corrections. Local open decisions include final α boundedness tolerance, CFL limit, pressure-residual tolerance, force/moment consistency tolerance and energy-diagnostic normalisation.

9.20 Requirements and Handoffs

Software shall emit the RES-VER-MET metrics for every run and tag each report as V1, V2 or V3. RES-DAT-CAS supplies observation bundles and calibration/validation split metadata; RES-GEO-SPEC supplies geometry identifiers; RES-NUM-SPEC supplies CFL, residual, boundedness and mass-conservation diagnostics. This handoff supports test implementation but does not itself execute validation.

9.21 Deliverable Completion Record

References

1. Qin X, Motley M, LeVeque R, Gonzalez F, and Mueller K. A Comparison of a Two-Dimensional Depth-Averaged Flow Model and a Three-Dimensional RANS Model for Predicting Tsunami Inundation and Fluid Forces. *Natural Hazards and Earth System Sciences* 2018; 18:2489–506. DOI: [10.5194/nhess-18-2489-2018](https://doi.org/10.5194/nhess-18-2489-2018)
2. Watanabe M, Arikawa T, Kihara N, Tsurudome C, Hosaka K, Kimura T, Hashimoto T, Ishihara F, Shikata T, Morikawa DS, Makino T, Asai M, Chida Y, Ohnishi Y, Marras S, Mukherjee A, Cajas JC, Houzeaux G, B D Paolo, Lara JL, Barajas G, Losada ÍJ, Hasebe M, Shigihara Y, Asai T, Ikeya T, Inoue S, Matsutomi H, Nakano Y, Okuda Y, Okuno S, Ooie T, Shoji G, and Tateno T. Validation of Tsunami Numerical Simulation Models for an Idealized Coastal Industrial Site. *Coastal Engineering Journal* 2022; 64:302–43. DOI: [10.1080/21664250.2022.2072611](https://doi.org/10.1080/21664250.2022.2072611)
3. Grilli ST, Harris JC, Tajalli-Bakhsh TST, Masterlark TL, Kyriakopoulos C, Kirby JT, and Shi F. Numerical Simulation of the 2011 Tohoku Tsunami Based on a New Transient FEM Co-Seismic Source: Comparison to Far- and Near-Field Observations. *Pure and Applied Geophysics* 2013; 170:1333–59. DOI: [10.1007/s00024-012-0528-y](https://doi.org/10.1007/s00024-012-0528-y)
4. Wei Y, Chamberlin CD, Titov VV, Tang L, and Bernard EN. Modeling of the 2011 Japan Tsunami: Lessons for Near-Field Forecast. *Pure and Applied Geophysics* 2013; 170:1309–31. DOI: [10.1007/s00024-012-0519-z](https://doi.org/10.1007/s00024-012-0519-z)